About this PDF

Built only from: **NOC:Natural Language Processing (NPTEL course book, weeks 6, 8)** — https://nptel.ac.in/courses/106105158

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Covers 6 of 14 syllabus sub-topics of this unit.

Partly covered in the source:

- WSD using Supervised

Not in the source (not covered in this PDF):

- Requirements for representation
- First-Order Logic
- Description Logics
- Syntax-Driven Semantic analysis
- Semantic attachments
- Relations between Senses
- Thematic Roles
- selectional restrictions

Word Senses

Definition

A word sense is a particular meaning that a word can have in a given context, and a word may have multiple such meanings that can be distinguished by the words that co-occur with it.

Explanation

Word-sense induction (WSI) discovers the different senses of a word by building a co-occurrence graph centered on the target word; words that belong to the same sense become tightly connected, forming separate clusters. To decide which sense is intended in a specific sentence, a *sense bag* (the set of features from a dictionary definition of a candidate sense) is compared with a *context bag* (the set of words surrounding the target word). The sense whose sense bag has the greatest overlap with the context bag is selected. Graph-based ranking methods such as PageRank can also be applied: each sense is a node, weighted edges are created based on overlap of sense definitions, and the sense with the highest rank is chosen.

Example

The word **compiler** had two senses in the period 1909-1953 (a person who compiles information, e.g., reporter, auditor) and acquired a new sense in 2002-2005 referring to a programming-language compiler, a sense absent in the earlier period.

Algorithm / Steps

1. Build a co-occurrence graph for the target word using a corpus.

2. Apply a graph-based clustering algorithm (e.g., Chinese Whispers) to obtain sense clusters.
3. For each candidate sense, construct a **sense bag** from dictionary definitions, example sentences, hyponyms, etc.
4. Construct a **context bag** from the words surrounding the target word in the sentence.
5. Compute the overlap between each sense bag and the context bag; select the sense with maximum overlap.
6. (Optional) Create weighted edges between sense nodes based on definition overlap, run a graph-ranking algorithm (e.g., PageRank), and choose the highest-scoring sense.

Key points

- Word-sense induction clusters co-occurring words to reveal distinct senses of a target word.
- Comparing sense clusters from different time periods can reveal novel senses that have emerged.
- The overlap between a sense bag and a context bag is a simple yet effective criterion for sense selection.
- Graph-based methods (clustering, PageRank) provide a flexible framework for modeling sense relationships.

Word Sense Disambiguation

Definition

Word Sense Disambiguation (WSD) is the task of determining which sense of an ambiguous word is invoked in a particular use, based on its context.

Explanation

WSD can be approached by exploiting the connections among the senses of words in a sentence. One method treats the problem as a graph where each possible sense of each word is a node; weighted edges are added between sense nodes according to the overlap of their dictionary definitions (e.g., using Lesk's method). A graph-based ranking algorithm such as PageRank is then applied, and the sense with the highest score for each word is selected, because it is connected to many other senses. Another simple approach builds a **sense bag** (words from a candidate sense definition) and a **context bag** (words from the surrounding context and definitions of context words) and chooses the sense whose bag has the maximum overlap with the context bag. A machine-learning alternative uses a Naive Bayes classifier: for each word, the sense that maximizes the probability given extracted contextual features is chosen.

Algorithm / Steps

1. For each ambiguous word, list all its possible senses (nodes).

2. Construct weighted edges between sense nodes by measuring overlap of their dictionary definitions (e.g., Lesk's method).
3. Apply a graph-based ranking algorithm (PageRank) to compute a score for each sense node.
4. For each word, select the sense with the highest PageRank score as the disambiguated sense.
5. (Alternative) Build a sense bag from the definition of each candidate sense and a context bag from surrounding words and their definitions; choose the sense with maximum overlap.
6. (Alternative) Train a Naive Bayes classifier using contextual features; for a given word, predict the sense with the highest posterior probability.

Key points

- WSD selects the appropriate sense of an ambiguous word by analysing its context.
- Graph-based methods connect sense nodes via definition overlap and rank them with PageRank, favoring senses linked to many others.
- Overlap-based methods compare sense bags with context bags and pick the maximum overlap.
- Naive Bayes treats sense selection as a classification problem using features extracted from the surrounding context.

WSD using Supervised

Partly covered in the source.

Definition

Word Sense Disambiguation (WSD) is the task of determining which sense of an ambiguous word is invoked in a particular usage, based on its context.

Explanation

The problem is addressed by examining the surrounding context of the target word. Various algorithms have been developed, including knowledge-base methods that rely on overlap techniques using resources such as machine-readable dictionaries (e.g., WordNet), as well as machine-learning approaches that can be supervised, unsupervised, or semi-supervised.

Key points

- WSD selects the appropriate sense of a word from multiple possible senses.
- Contextual information surrounding the word is essential for disambiguation.
- Approaches include knowledge-base overlap methods and machine-learning methods (supervised, unsupervised, semi-supervised).

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Dictionary & Thesaurus

Definition

A dictionary provides sense definitions, example sentences, hyponyms and other lexical features for each word, while a thesaurus groups words into categories that reflect their semantic relations. These resources are used to build sense bags and context bags for word-sense disambiguation.

Explanation

The disambiguation process extracts features from the dictionary (definitions, examples, hyponyms) to form a **sense bag** for each possible meaning of an ambiguous word. Simultaneously, words surrounding the target word contribute their own dictionary definitions to create a **context bag**. Overlap between a sense bag and the context bag indicates the most appropriate sense. A simple thesaurus-based approach adds a score of 1 for each context word whose thesaurus category matches the category of a candidate sense; the sense with the highest total score is selected.

Example

In the sentence “the money in this **bank** fetches interest of 8 percent per annum,” the ambiguous word *bank* has two senses: finance (safe for money) and location (river bank). Context words *money*, *interest*, and *annum* each belong to the finance category. Adding a score of 1 for each matching context word gives the finance sense a total score of 3 and the location sense a score of 0, so the word is interpreted as a financial institution.

Algorithm / Steps

1. Identify the ambiguous target word and list all its possible senses.
2. For each sense, note its thesaurus category (e.g., finance, location).
3. Collect context words from the sentence (e.g., *money*, *interest*, *annum*).
4. For each context word, determine its thesaurus category.
5. Add 1 to the score of a sense if a context word’s category matches that sense’s category; otherwise add 0.
6. Sum the scores for each sense and select the sense with the highest total.

Key points

- Dictionary features (definitions, examples, hyponyms) are used to construct sense bags.
- Thesaurus categories enable a simple scoring method for sense selection.
- The sense with the maximum overlap or highest score is chosen as the correct interpretation.

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Bootstrapping methods

Definition

Bootstrapping based methods are one of the approaches that can be employed to solve the word sense disambiguation problem, where the goal is to determine the appropriate sense of an ambiguous word in a given context.

Key points

- They are listed among several techniques for word sense disambiguation, alongside knowledge-base, machine-learning, and unsupervised methods.

Word Similarity using Thesaurus and Distributional methods.

Definition

Word similarity can be measured by using a lexical thesaurus such as WordNet or by building a distributional thesaurus from corpus co-occurrence statistics. The former relies on manually encoded semantic relations, while the latter derives similarity from the contexts in which words appear.

Explanation

A distributional thesaurus is created by counting co-occurrence of words in a large corpus and representing each word by its context vector. Words that share many contexts obtain high similarity scores. By applying word sense induction on these vectors, distinct sense clusters for a word can be discovered, allowing detection of new senses over time. Lesk's method computes similarity between sense definitions by counting overlapping words; these similarity scores become edge weights in a graph that connects sense nodes from different words. A graph-based ranking algorithm such as PageRank then highlights senses that are strongly connected to many other senses, providing a way to select the most appropriate sense for each word in a sentence.

Example

Using a distributional thesaurus built from two time periods, the word **register** was found to have an older sense cluster containing "dictionaries, directories, libraries" and a newer sense cluster containing "peripherals, processor, circuits, workstations". The new sense was detected because the word co-occurred with hardware-related terms in the later corpus, forming a distinct cluster.

Algorithm / Steps

1. Choose two time points *t1* and *t2* and extract corpus slices for each period.
2. Compute co-occurrence statistics for all words in each slice and construct a distributional thesaurus.
3. Perform word sense induction on the thesaurus to obtain sense clusters *s1, s2, …* for each word at *t1* and *t2*.
4. Compare sense clusters across the two time points to identify senses that appear only in *t2* (new senses).

5. For a given sentence, list all possible senses of each word and build a graph where nodes are senses.
6. Compute edge weights between sense nodes using Lesk's overlap similarity of their definitions.
7. Apply a PageRank-like graph ranking algorithm to score each sense node.
8. Select, for each word, the sense with the highest score as the disambiguated sense.

Key points

- Distributional thesaurus derives similarity from shared contexts, enabling detection of emerging word senses.
- Lesk's overlap method provides similarity scores between sense definitions, which are used as edge weights in a graph.
- Graph-based ranking (e.g., PageRank) favors senses that are highly connected to other senses, aiding word sense disambiguation.
- Comparing sense clusters from different time periods reveals sense change, such as the shift of "register" toward a computing sense.

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Changes made by AcademicsGen: the source text was rephrased, reorganised and grammar-corrected into study material; nothing added.

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